NAME

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PROJECT

LOGISTICS PROJECT

CLASS

POST-MENTORSHIP DATA CREW: PROJECT 2 PRACTICE(P2P)

DATE

20th OF MARCH 2026.

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EXECUTIVE SUMMARY

PURPOSE OF THE REPORT

The purpose of the report is to provide a comprehensive analysis of logistics performance between 2020 and 2024, focusing on shipment efficiency, customer satisfaction, and carrier performance across multiple transport modes and regions. The goal of this project is to deliver a holistic view of the company's logistics performance by leveraging key performance indicators such as delivery time, shipment value, and shipment status.

The analysis will aim to reveal underlying trends and performance fluctuations across different transport modes and regions, helping to pinpoint areas of strength and those requiring improvement. Ultimately, the project seeks to provide actionable insights that will enable stakeholders to make informed decisions, enhance operational efficiency, streamline delivery processes, and improve overall customer satisfaction.

SCOPE AND AUDIENCE

The intended readers are senior executives, logistics managers, and analysts responsible for operational planning, resource allocation, and customer experience strategies.

KEY FINDINGS

- Shipment completion rates show inconsistencies, with delays in high-priority deliveries.
- Regional disparities in delivery times and shipment value highlight inefficiencies in route planning.
- Customer satisfaction varies, strongly influenced by payment status and delivery delays.
- Carrier performance ratings reveal opportunities to optimize partnerships.

PROJECT BACKGROUND

BUSINESS PROBLEM STATEMENT

The logistics company is facing challenges in maintaining delivery efficiency, ensuring customer satisfaction, and optimizing shipment performance. Despite growing demand, inconsistencies in shipment completion rates, delays in high-priority deliveries, and varied customer feedback have raised concerns. Specific transport modes such as Motorbikes and Lorries show weaker performance, while regional disparities in delivery times and shipment value highlight inefficiencies in route planning and carrier management.

SCOPE OF ANALYSIS(2020-2024)

The dataset spans four years, covering multiple transport modes (Jets, Bus, Lorry, Motorbike) and regions (Lagos, Abuja, Port Harcourt, Accra, Kano, Ibadan, etc.). It includes shipment status, delivery times, shipment values, customer satisfaction scores, and carrier performance ratings. This scope allows for both trend analysis over time and comparative analysis across modes and regions.

PROJECT OBJECTIVES

1. Quantify completed and in-progress shipments across multiple transport modes (Jet, Bus, Lorry, Motorbike).
2. Evaluate shipment value trends and the financial performance of logistics services.
3. Analyze shipment efficiency and delivery performance across key regions (Lagos, Abuja, Port Harcourt, etc.).
4. Measure customer satisfaction levels, identify drivers of positive or negative experiences, and examine the impact of shipment delays or payment status on ratings.
5. Recommend improvements to enhance shipment delivery timelines, optimize transport mode deployment, and increase overall satisfaction.

SOURCE DATA

- **DATA ORIGIN AND FORMAT**
  - Source: Excel shipment file provided by the logistics company
  - Format: Tabular dataset with one row per shipment
  - Timeframe: 2020–2024

  This verifies that each record represents a single shipment transaction which includes shipment details, customer attributes, carrier information, and performance metrics.

- **COLUMNS DESCRIPTION(DATA DICTIONARY)**

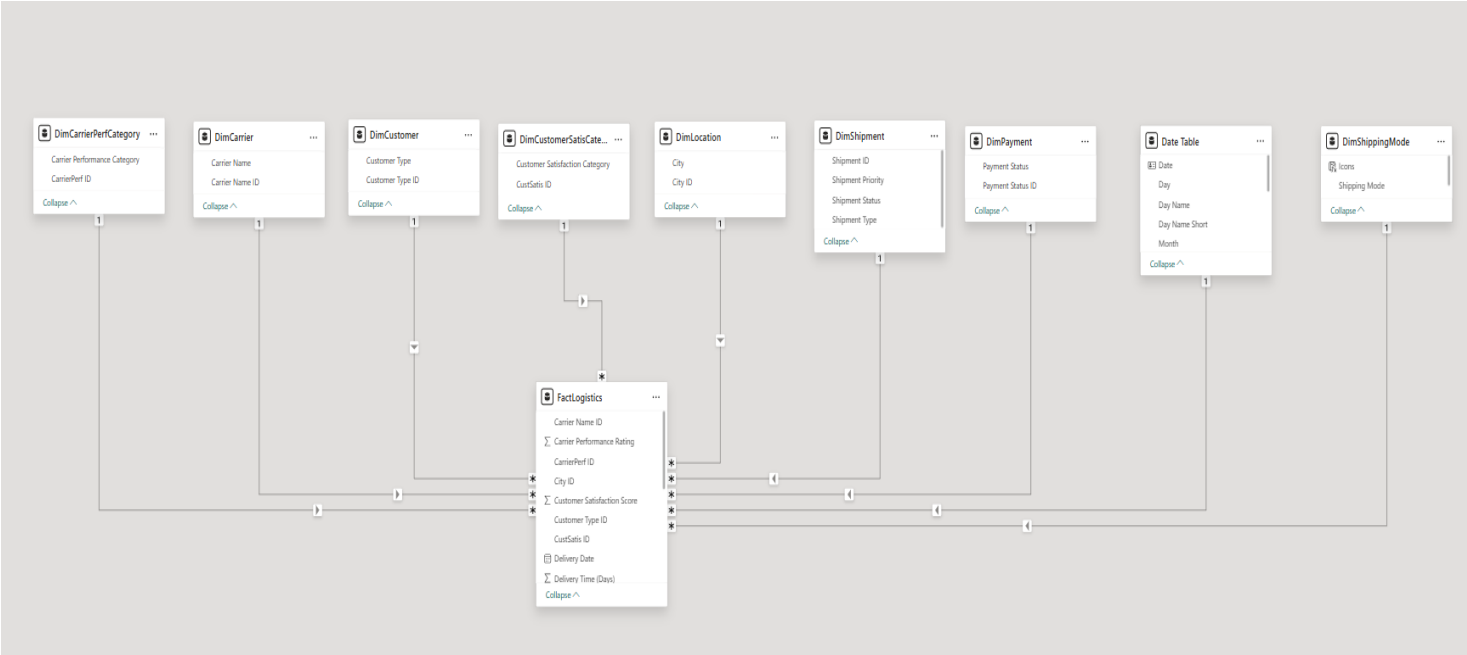
Column Name	**Description**
Shipment ID	Unique identifier for each shipment.
Shipment Status	Current status of the shipment (e.g., Completed, Ready for Pickup, In Transit, Not Completed).
Order Date	The date when the shipment was ordered.
Delivery Date	The actual delivery date of the shipment.
Shipping Mode	Mode of transport used (Jets, Bus, Lorry, Motorbike).
Carrier Name	Name of the logistics carrier handling the shipment.
Shipment Weight (lbs)	Weight of the shipment in pounds.
Delivery Time (Days)	Number of days it took from order to delivery.
Delivery Location	The final destination of the shipment (city and country).
Shipment Cost/Weight ($)	Cost per unit weight of the shipment.
Loading Time (hrs)	Time taken to load the shipment, in hours.
Loading Weight (lbs)	Weight of shipment at loading, may vary slightly from shipment weight.
Estimated Delivery Date	Forecasted delivery date at the time of order.
Tracking Number	Unique tracking number assigned to the shipment.
Shipment Type	Classification such as Standard, Express, or Urgent.
Customer Type	Type of customer (Business, Retail, International).
Payment Status	Indicates whether the payment is Paid, Pending, or Overdue.
Shipment Value ($)	Monetary value of the shipment.
Shipment Priority	Priority level (High, Medium, Low).

Carrier Performance Rating	Carrier service rating (scale of 1–5).
Delivery Window	Preferred delivery window (e.g., 9 AM – 12 PM).
Customer Satisfaction Score	Satisfaction rating (scale of 1–10).

DATA CLEANING AND TRANSFORMATION STEPS

- **Applied Steps in Power Query**
  - **Source** – Imported logistics Excel file
  - **Promoted Headers** – Ensured column names were correctly recognized
  - **Changed Type** – Converted columns to appropriate data types (dates, numbers, text)
  - **Added Custom Columns** – Derived new fields (e.g., Delivery Duration = Delivery Date – Order Date)
  - **Split Column by Delimiter** – Separated composite fields where necessary
  - **Renamed Columns** – Standardized naming conventions for clarity
  - **Removed Columns / Removed Other Columns** – Dropped irrelevant fields
  - **Removed Duplicates** – Ensured unique table IDs
  - **Added Index** – Created row identifiers for tracking
  - **Reordered Columns** – Organized for consistency
- **Merging Queries with Dimension Tables**
  - Linked shipment records to **DimCarrier** using Carrier Name ID
  - Linked shipment records to **DimCustomer** using Customer Type ID
  - Linked shipment records to **DimLocation** using City ID
  - Linked shipment records to **DimPayment** using Payment Status ID
  - Linked shipment records to **DimShipment** using Shipment ID and Priority
  - Linked shipment records to **DimShippingMode** using Shipping Mode
  - Linked shipment records to **DimCustomerSatisfCate** using Satisfaction Category ID
  - Linked shipment records to **DimCarrierPerfCategory** using Carrier Performance ID
- **Outcome of Cleaning & Merging**
  - A clean, consistent dataset ready for modeling
  - Fact table (FactLogistics) enriched with dimension IDs for relationships
  - Ensures accurate filtering, slicing, and aggregation in dashboards

DATA MODEL



- **Schema Overview**
  - Design: Star schema
  - Central fact table: **FactLogistics**
  - Dimension tables: Carrier, Customer, Location, Payment, Shipment, Shipping Mode, Date, Satisfaction Category, Performance Category
  - Purpose: Ensure efficient filtering, slicing, and aggregation across multiple perspectives (time, mode, region, customer, carrier)
- **Fact Table: FactLogistics**
  - Contains measurable data such as shipment value, delivery time, customer satisfaction score, and carrier performance rating
  - Includes foreign keys linking to dimension tables (Carrier ID, City ID, Customer Type ID, Payment Status ID, etc.)
  - One row per shipment record
- **Dimension Tables**
  - **DimCarrier** – Carrier Name, Carrier ID
  - **DimCarrierPerfCategory** – Carrier Performance Category, Perf ID
  - **DimCustomer** – Customer Type, Customer Type ID
  - **DimCustomerSatisfCate** – Satisfaction Category, CustSatis ID
  - **DimLocation** – City, City ID
  - **DimShipment** – Shipment ID, Priority, Status, Type
  - **DimPayment** – Payment Status, Payment Status ID
  - **DimShippingMode** – Shipping Mode, Icons, Shipping mode ID
  - **Date Table** – Date, Day, Month, Year (supports time intelligence)

- **Relationships**
  - FactLogistics[Carrier Name ID] → DimCarrier[Carrier Name ID]
  - FactLogistics[City ID] → DimLocation[City ID]
  - FactLogistics[Customer Type ID] → DimCustomer[Customer Type ID]
  - FactLogistics[Shipment ID] → DimShipment[Shipment ID]
  - FactLogistics[Shipping mode ID] → DimShippingMode[Shipping Mode ID]
  - FactLogistics[Payment Status ID] → DimPayment[Payment Status ID]
  - FactLogistics[CustSatis ID] → DimCustomerSatisfCate[CustSatis ID]
  - FactLogistics[CarrierPerf ID] → DimCarrierPerfCategory[CarrierPerf ID]
  - FactLogistics[Date] → Date Table[Date]
- **Usage in Dashboards**
  - **Page 1 (Overview)** – FactLogistics + Date Table for KPIs and trends
  - **Page 2 (Shipping Mode Analysis)** – FactLogistics + DimShippingMode for mode comparisons
  - **Page 3 (Customer Experience)** – FactLogistics + DimCustomer, DimPayment, DimCustomerSatisfCate for satisfaction and payment breakdowns

DAX MEASURES

- **PURPOSE**

To provide a clear record of all calculated measures used in the dashboards, including their formulas and business meaning. This ensures reproducibility and helps stakeholders understand how KPIs are derived.

MEASURE NAME	DAX FORMULA	PURPOSE/BUSINESS MEANING
% C_Early	`VAR Early = CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "Completed", FactLogistics[Delivery Date] <= FactLogistics[Estimated Delivery Date]) RETURN DIVIDE(Early, [S_Completed], 0)`	Percentage of shipments delivered on or before the estimated delivery date. Measures punctuality.
% C_Late	`VAR Late = CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "Completed", FactLogistics[Delivery Date] > FactLogistics[Estimated Delivery Date]) RETURN DIVIDE(Late, [S_Completed], 0)`	Percentage of shipments delivered after the estimated delivery date. Highlights delays.
%Completed	`DIVIDE([S_Completed], [Total Shipment], 0)`	Share of shipments marked as completed out of total shipments. Overall completion rate.
%In Transit	`VAR Transit = CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "In Transit") VAR PercentTransit = DIVIDE(Transit, [Total Shipment], 0) RETURN PercentTransit`	Share of shipments currently in transit. Indicates ongoing deliveries.
%Ready for pickup	`DIVIDE([S_Ready for Pickup], [Total Shipment], 0)`	Share of shipments awaiting pickup. Shows backlog at pickup stage.
%Not completed	`DIVIDE([S_Not Completed], [Total Shipment], 0)`	Share of shipments not completed. Identifies operational inefficiencies.
Avg Delivery Day	`AVERAGE(FactLogistics[Delivery Time (Days)])`	Average number of days taken to deliver

		shipments. Core efficiency KPI.
Avg Satisfaction Score	`AVERAGE(FactLogistics[Customer Satisfaction Score])`	Average customer satisfaction rating. Reflects service quality.
S_Completed	`CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "Completed")`	Count of shipments completed.
S_IN Progress	`CALCULATE( COUNTROWS(FactLogistics), DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}`	Count of shipments still in progress (in transit, not completed, ready for pickup).
S_In Transit	`CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "In Transit")`	Count of shipments currently in transit.
S_Not Completed	`CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "Not Completed")`	Count of shipments not completed.
S_Ready for Pickup	`CALCULATE(COUNTROWS(FactLogistics), DimShipment[Shipment Status] = "Ready for Pickup")`	Count of shipments awaiting pickup.
Shipment Value	`SUM(FactLogistics[Shipment Value ($)])`	Total number of shipments recorded.
Total Shipment	`COUNT(FactLogistics[Shipment ID])`	Total monetary value of shipments.
% Overdue	`VAR Overdue = CALCULATE(_Shipment[Shipment Value], DimPayment[Payment Status] = "Overdue") RETURN DIVIDE(Overdue, _Shipment[Shipment Value], 0)`	Percentage of shipment value tied to overdue payments. Indicates financial risk.
% Paid	`VAR Paid = CALCULATE(_Shipment[Shipment Value], DimPayment[Payment Status] = "Paid") RETURN DIVIDE(Paid, _Shipment[Shipment Value], 0)`	Percentage of shipment value successfully paid. Shows revenue realization.
% Pending	`VAR Pending = CALCULATE(_Shipment[Shipment Value], DimPayment[Payment Status] = "Pending") RETURN DIVIDE(Pending, _Shipment[Shipment Value], 0)`	Percentage of shipment value still pending. Highlights cash flow delays.
Location_Title_Bus	`VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH( TRUE(),`	Dynamic titles for visuals showing completed vs in-progress shipments by city for each transport mode.

	StatusSelected = "Completed", "Completed Shipments by Bus In Each City", StatusSelected = "In Progress", "In Progress Shipments by Bus In Each City")	
Location_Title_Jets	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Shipments by Jets In Each City", StatusSelected = "In Progress", "In Progress Shipments by Jets In Each City")	Dynamic titles for visuals showing completed vs in-progress shipments by city for each transport mode.
Location_Title_Lorry	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Shipments by Lorry In Each City", StatusSelected = "In Progress", "In Progress Shipments by Lorry In Each City")	Dynamic titles for visuals showing completed vs in-progress shipments by city for each transport mode.
Location_Title_Motorbike	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Shipments by Motorbike In Each City", StatusSelected = "In Progress", "In Progress Shipments by Motorbike In Each City")	Dynamic titles for visuals showing completed vs in-progress shipments by city for each transport mode.
Mode_bus	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics),	Counts shipments by mode, filtered by status (completed or in-progress).

	DimShippingMode[Shipping Mode] = "Bus", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
Mode_Jets	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Jets", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by mode, filtered by status (completed or in-progress).
Mode_Lorry	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Lorry", SWITCH(TRUE(),	Counts shipments by mode, filtered by status (completed or in-progress).

	Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
Mode_Motorbike	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Motorbike", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by mode, filtered by status (completed or in-progress).
P_High_Bus	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Bus", DimShipment[Shipment Priority] = "High", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_High_Jets	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Jets", DimShipment[Shipment Priority] = "High", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_High_Lorry	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Lorry", DimShipment[Shipment Priority] = "High", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_High_Motorbike	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Motorbike", DimShipment[Shipment Priority] = "High", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_Low_Bus	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Bus", DimShipment[Shipment Priority] = "Low", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_Low_Jets	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Jets", DimShipment[Shipment Priority] = "Low", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_Low_Lorry	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Lorry", DimShipment[Shipment Priority] = "Low", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_Low_Motorbike	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Motorbike", DimShipment[Shipment Priority] = "Low", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_Medium_Bus	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Bus", DimShipment[Shipment Priority] = "Medium", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_Medium_Jets	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Jets", DimShipment[Shipment Priority] = "Medium", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_Medium_Lorry	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Lorry", DimShipment[Shipment Priority] = "Medium", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed",	Counts shipments by priority level for each mode, filtered by status.

	InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	
P_Medium_Motorbike	VAR StatusSelected =SELECTEDVALUE('Shipping Status'[Status], "Completed") VAR Completed = StatusSelected = "Completed" VAR InProgress = StatusSelected = "In Progress" RETURN CALCULATE(COUNTROWS(FactLogistics), DimShippingMode[Shipping Mode] = "Motorbike", DimShipment[Shipment Priority] = "Medium", SWITCH(TRUE(), Completed, DimShipment[Shipment Status] = "Completed", InProgress, DimShipment[Shipment Status] IN {"In Transit", "Not Completed", "Ready for Pickup"}))	Counts shipments by priority level for each mode, filtered by status.
P_Title_Bus	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Priority Shipments by Bus", StatusSelected = "In Progress", "In Progress Priority Shipments by Bus")	Dynamic titles for visuals showing priority shipments by mode.
P_Title_Jets	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Priority Shipments by Jets",	Dynamic titles for visuals showing priority shipments by mode.

	StatusSelected = "In Progress", "In Progress Priority Shipments by Jets")	
P_Title_Lorry	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Priority Shipments by Lorry", StatusSelected = "In Progress", "In Progress Priority Shipments by Lorry")	Dynamic titles for visuals showing priority shipments by mode.
P_Title_Motorbike	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Completed Priority Shipments by Motorbike", StatusSelected = "In Progress", "In Progress Priority Shipments by Motorbike")	Dynamic titles for visuals showing priority shipments by mode..
Title_Bus	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Shipments Completed by Bus", StatusSelected = "In Progress", "Shipments Expected to be Delivered by Bus")	Dynamic titles for visuals showing shipments completed vs expected by mode.
Title_Jets	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Shipments Completed by Jets", StatusSelected = "In Progress", "Shipments Expected to be Delivered by Jets")	Dynamic titles for visuals showing shipments completed vs expected by mode.
Title_Lorry	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(	Dynamic titles for visuals showing shipments completed vs expected by mode.

	TRUE(), StatusSelected = "Completed", "Shipments Completed by Lorry", StatusSelected = "In Progress", "Shipments Expected to be Delivered by Lorry")	
Title_Motorbike	VAR StatusSelected = SELECTEDVALUE('Shipping Status'[Status], "Completed") RETURN SWITCH(TRUE(), StatusSelected = "Completed", "Shipments Completed by MotorBike", StatusSelected = "In Progress", "Shipments Expected to be Delivered by MotorBike")	Dynamic titles for visuals showing shipments completed vs expected by mode.
MOM S_Completed	VAR PMCompleted = CALCULATE([S_Completed], PREVIOUSMONTH('Date Table'[Date])) VAR MoM = DIVIDE([S_Completed], PMCompleted) -1 RETURN IF(MoM > 0, "▲", "▼") & FORMAT(MoM, "0%")	Month-over-month change in completed shipments (% increase or decrease).
MOM S_In Progress	VAR PMProgress = CALCULATE([S_IN Progress], PREVIOUSMONTH('Date Table'[Date])) VAR MoM = DIVIDE([S_IN Progress], PMProgress) -1 RETURN IF(MoM > 0, "▲", "▼") & FORMAT(MoM, "0%")	Month-over-month change in in-progress shipments.
MOM Shipment_Value	VAR PMValue = CALCULATE([Shipment Value], PREVIOUSMONTH('Date Table'[Date])) VAR MoM = DIVIDE([Shipment Value], PMValue) -1 RETURN IF(MoM > 0, "▲", "▼") & FORMAT(MoM, "0%")	Month-over-month change in shipment value.
YOY Completed	VAR SPLY_Completed = CALCULATE([S_Completed], SAMEPERIODLASTYEAR('Date Table'[Date])) VAR YOY = DIVIDE([S_Completed], SPLY_Completed) -1 RETURN IF(YOY > 0, "▲", "▼") & FORMAT(YOY, "0%")	Year-over-year change in completed shipments.

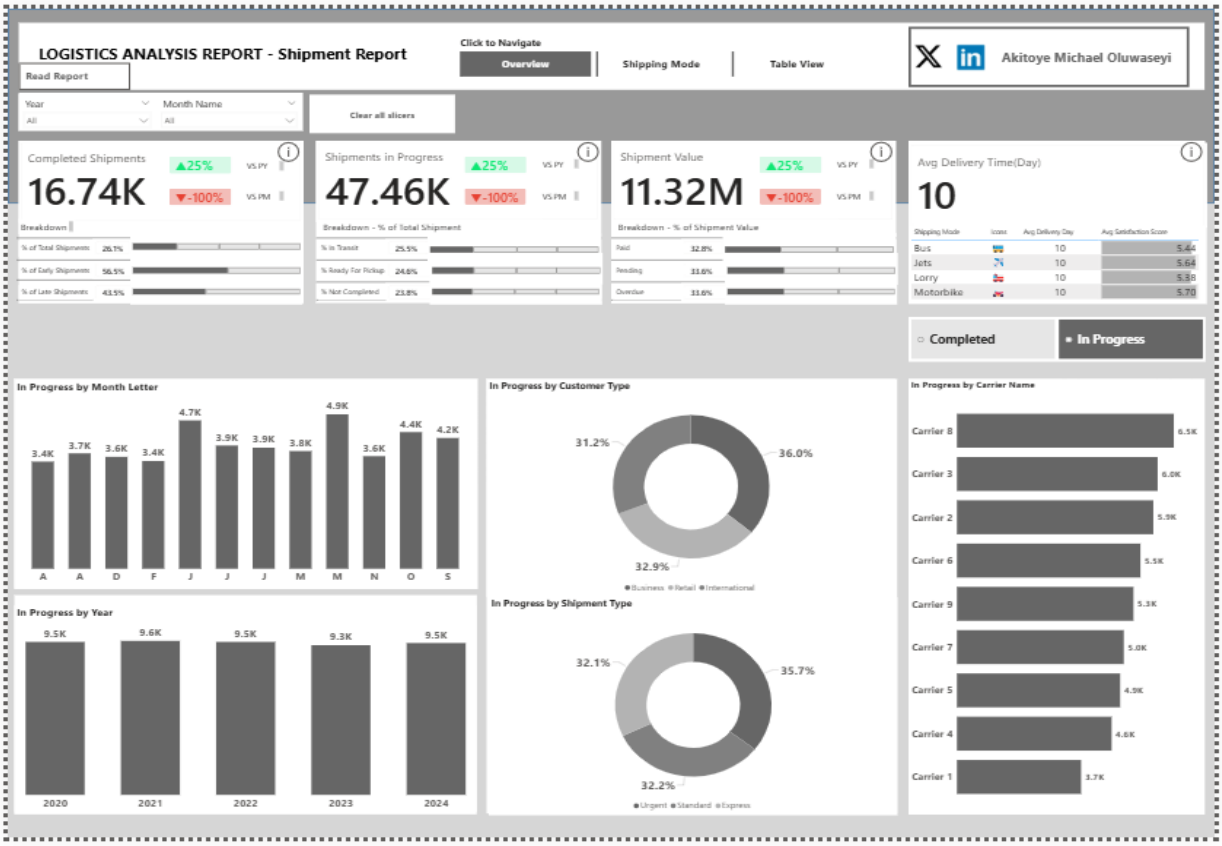
YOY S_In Progress	VAR SPLY_IN_Progress = CALCULATE([S_IN Progress], SAMEPERIODLASTYEAR('Date Table'[Date])) VAR YOY = DIVIDE([S_IN Progress], SPLY_IN_Progress) -1 RETURN IF(YOY > 0, "▲", "▼") & FORMAT(YOY, "0%")	Year-over-year change in in-progress shipments.
YOY Shipment_Value	VAR SValue = CALCULATE([Shipment Value], SAMEPERIODLASTYEAR('Date Table'[Date])) VAR YOY = DIVIDE([Shipment Value], SValue) -1 RETURN IF(YOY > 0, "▲", "▼") & FORMAT(YOY, "0%")	Year-over-year change in shipment value.

Date Table DAX

```
= ADDCOLUMNS(
CALENDARAUTO(),
"Year", YEAR([Date]),
"Qtr Name", FORMAT([Date], "\QQ"),
"Quarter", QUARTER([Date]),
"Month", MONTH([Date]),
"Month Name", FORMAT([Date], "mmmm"),
"Month Name Short", FORMAT([Date], "mmm"),
"Day", DAY([Date]),
"WeekDay", WEEKDAY([Date]),
"Day Name", FORMAT([Date], "dddd"),
"Day Name Short", FORMAT([Date], "ddd"),
"Week Num", WEEKNUM([Date]),
"Weekend", IF(WEEKDAY([Date]) IN {1,7} , "Weekend", "Weekday")
```

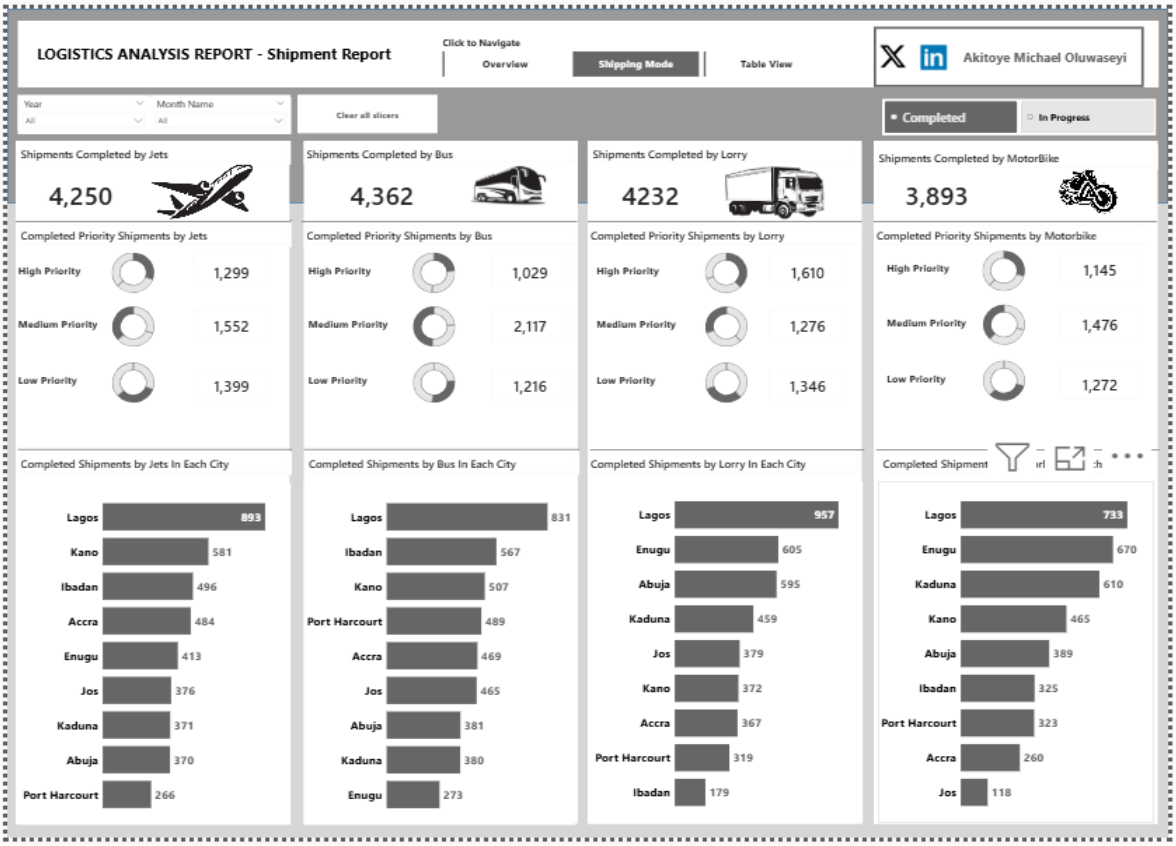
DASHBOARD SUMMARY

📊 PAGE 1 – OVERVIEW



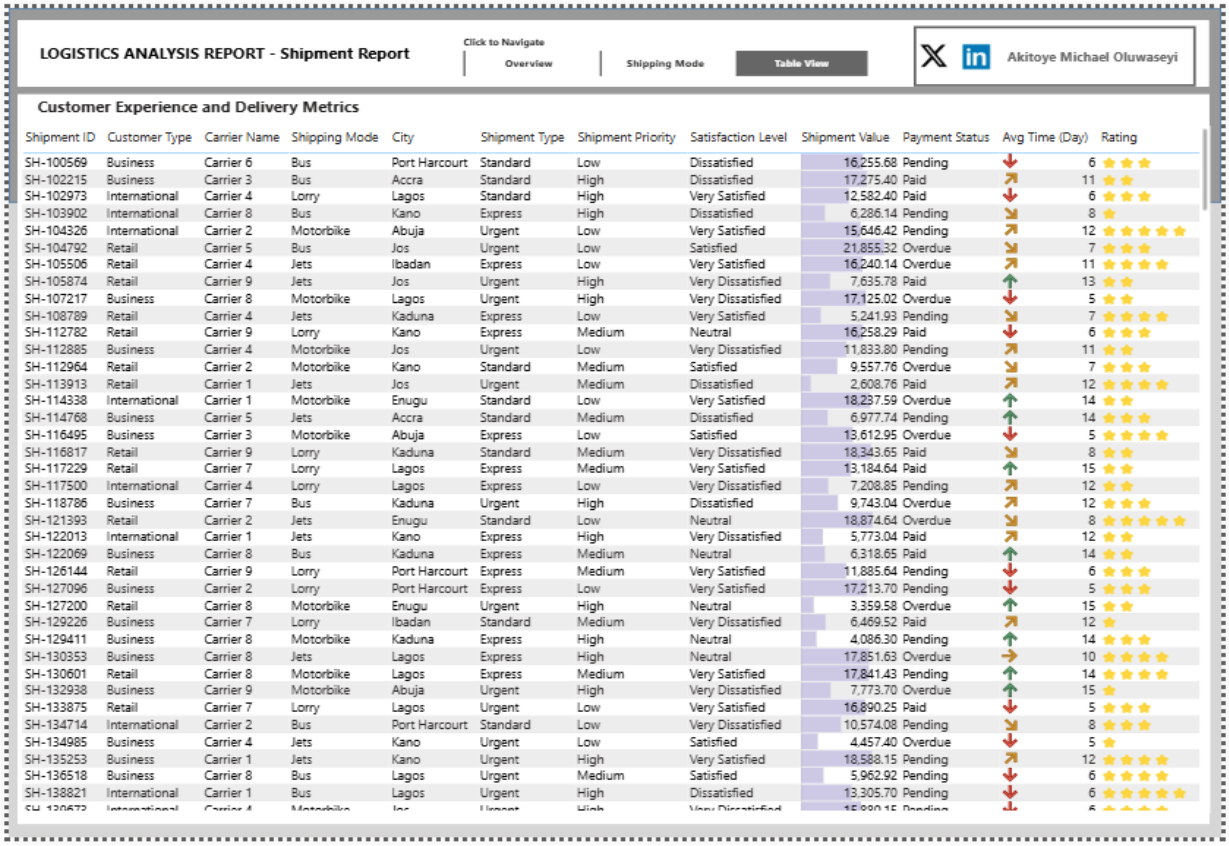
- **Purpose:** Provide executives with a high-level snapshot of logistics performance.
- **KPIs & Measures Used:**
  - % Completed, % In Transit, % Not Completed, % Ready for Pickup
  - Shipment Value, Avg Delivery Day, Avg Satisfaction Score
  - MOM & YOY measures (Completed, In Progress, Shipment Value)
- **Visuals:**
  - KPI cards for shipment status percentages
  - Line chart showing shipment trends over time
  - Bar chart comparing carrier performance ratings
  - Trend indicators (▲ ▼) for MoM and YoY changes
- **Insights Delivered:**
  - Overall completion rates and delays
  - Month-over-month and year-over-year shipment growth or decline
  - Carrier reliability comparison

🚚 PAGE 2 – SHIPPING MODE ANALYSIS



- **Purpose:** Break down shipment volumes and performance by transport mode.
- **KPIs & Measures Used:**
  - Mode_Bus, Mode_Jets, Mode_Lorry, Mode_Motorbike
  - Priority measures (P_High, P_Medium, P_Low for each mode)
  - Location_Title measures for dynamic city-level analysis
- **Visuals:**
  - Stacked bar charts showing completed vs in-progress shipments by mode
  - Tables showing shipment counts by city and mode
  - Priority distribution visuals (High, Medium, Low)
- **Insights Delivered:**
  - Identification of bottlenecks in Motorbike and Lorry shipments
  - Regional disparities in shipment volumes and completion rates
  - Demand patterns by priority level and transport mode

👥 PAGE 3- CUSTOMER EXPERIENCE AND DELIVERY METRICS



LOGISTICS ANALYSIS REPORT - Shipment Report

Click to Navigate | Overview | Shipping Mode | Table View

Akitoye Michael Oluwaseyi

Customer Experience and Delivery Metrics

Shipment ID	Customer Type	Carrier Name	Shipping Mode	City	Shipment Type	Shipment Priority	Satisfaction Level	Shipment Value	Payment Status	Avg Time (Day)	Rating
SH-100569	Business	Carrier 6	Bus	Port Harcourt	Standard	Low	Dissatisfied	16,255.68	Pending	6	★★★
SH-102215	Business	Carrier 3	Bus	Accra	Standard	High	Dissatisfied	17,275.40	Paid	11	★★
SH-102973	International	Carrier 4	Lorry	Lagos	Standard	High	Very Satisfied	12,582.40	Paid	6	★★★
SH-103902	International	Carrier 8	Bus	Kano	Express	High	Dissatisfied	6,286.14	Pending	8	★
SH-104326	International	Carrier 2	Motorbike	Abuja	Urgent	Low	Very Satisfied	15,646.42	Pending	12	★★★★★
SH-104792	Retail	Carrier 5	Bus	Jos	Urgent	Low	Satisfied	21,855.32	Overdue	7	★★★
SH-105506	Retail	Carrier 4	Jets	Ibadan	Express	Low	Very Satisfied	16,240.14	Overdue	11	★★★★
SH-105874	Retail	Carrier 9	Jets	Jos	Urgent	High	Very Dissatisfied	7,635.78	Paid	13	★★
SH-107217	Business	Carrier 8	Motorbike	Lagos	Urgent	High	Very Dissatisfied	17,125.02	Overdue	5	★★
SH-108789	Retail	Carrier 4	Jets	Kaduna	Express	Low	Very Satisfied	5,241.93	Pending	7	★★★★
SH-112782	Retail	Carrier 9	Lorry	Kano	Express	Medium	Neutral	16,258.29	Paid	6	★★★
SH-112885	Business	Carrier 4	Motorbike	Jos	Urgent	Low	Very Dissatisfied	11,833.80	Pending	11	★★
SH-112964	Retail	Carrier 2	Motorbike	Kano	Standard	Medium	Satisfied	9,557.76	Overdue	7	★★★
SH-113913	Retail	Carrier 1	Jets	Jos	Urgent	Medium	Dissatisfied	2,608.76	Paid	12	★★★★
SH-114338	International	Carrier 1	Motorbike	Enugu	Standard	Low	Very Satisfied	18,237.59	Overdue	14	★★
SH-114768	Business	Carrier 5	Jets	Accra	Standard	Medium	Dissatisfied	6,977.74	Pending	14	★★★
SH-116495	Business	Carrier 3	Motorbike	Abuja	Express	Low	Satisfied	13,612.95	Overdue	5	★★★★
SH-116817	Retail	Carrier 9	Lorry	Kaduna	Standard	Medium	Very Dissatisfied	18,343.65	Paid	8	★★
SH-117229	Retail	Carrier 7	Lorry	Lagos	Express	Medium	Very Satisfied	13,184.64	Paid	15	★★
SH-117500	International	Carrier 4	Lorry	Lagos	Express	Low	Very Dissatisfied	7,208.85	Pending	12	★★
SH-118786	Business	Carrier 7	Bus	Kaduna	Urgent	High	Dissatisfied	9,743.04	Overdue	12	★★★
SH-121393	Retail	Carrier 2	Jets	Enugu	Standard	Low	Neutral	18,874.64	Overdue	8	★★★★★
SH-122013	International	Carrier 1	Jets	Kano	Express	High	Very Dissatisfied	5,773.04	Paid	12	★★
SH-122069	Business	Carrier 8	Bus	Kaduna	Express	Medium	Neutral	6,318.65	Paid	14	★★
SH-126144	Retail	Carrier 9	Lorry	Port Harcourt	Express	Medium	Very Satisfied	11,885.64	Pending	6	★★★
SH-127096	Business	Carrier 2	Lorry	Port Harcourt	Express	Low	Very Satisfied	17,213.70	Pending	5	★★★
SH-127200	Retail	Carrier 8	Motorbike	Enugu	Urgent	High	Neutral	3,359.58	Overdue	15	★★
SH-129226	Business	Carrier 7	Lorry	Ibadan	Standard	Medium	Very Dissatisfied	6,469.52	Paid	12	★
SH-129411	Business	Carrier 8	Motorbike	Kaduna	Express	High	Neutral	4,086.30	Pending	14	★★★
SH-130353	Business	Carrier 8	Jets	Lagos	Express	High	Neutral	17,851.63	Overdue	10	★★★★
SH-130601	Retail	Carrier 8	Motorbike	Lagos	Express	Medium	Very Satisfied	17,841.43	Pending	14	★★★★
SH-132938	Business	Carrier 9	Motorbike	Abuja	Urgent	High	Very Dissatisfied	7,773.70	Overdue	15	★
SH-133875	Retail	Carrier 7	Lorry	Lagos	Urgent	Low	Very Satisfied	16,890.25	Paid	5	★★★
SH-134714	International	Carrier 2	Bus	Port Harcourt	Standard	Low	Very Dissatisfied	10,574.08	Pending	8	★★★
SH-134985	Business	Carrier 4	Jets	Kano	Urgent	Low	Satisfied	4,457.40	Overdue	5	★
SH-135253	Business	Carrier 1	Jets	Kano	Urgent	High	Very Satisfied	18,588.15	Pending	12	★★★★
SH-136518	Business	Carrier 8	Bus	Lagos	Urgent	Medium	Satisfied	5,962.92	Pending	6	★★★★
SH-138821	International	Carrier 1	Bus	Lagos	Urgent	High	Dissatisfied	13,305.70	Pending	6	★★★★★

- **Purpose:** Evaluate customer satisfaction, payment status, and delivery efficiency.

INSIGHTS AND RECOMMENDATION

Page 1: Overview of Shipment Status

Key Insights:

Completed Shipments:

- Total completed shipments stand at 16.74K, with a 25% increase compared to the previous year.
- The value of completed shipments is $11.32M, with a 25% increase from the previous year.
- The average shipment time is 9.7 days, with Jets having the fastest delivery time at 10.1 days, and Motorbikes being the slowest at 9.9 days.

In Progress Shipments:

- 47.46K shipments are still in progress, with a 25% increase compared to last year.
- High priority shipments account for the highest proportion of in-progress deliveries.

Shipment Trend Analysis:

- The report also highlights trends over the months and years, giving insights into the growth and challenges in shipment deliveries.

Page 2: Shipment Forecasting by Shipping Mode

Key Insights:

Jet Shipments:

- 11,775 shipments are expected to be delivered by Jet, with High priority shipments accounting for 4,002 of the deliveries.

Bus Shipments:

- A total of 10,861 shipments are expected to be delivered by Bus, with Medium priority shipments accounting for 3,602 deliveries.

Lorry Shipments:

- 12,604 shipments are expected to be delivered by Lorry, with 4,430 being Low priority shipments.

Motorbike Shipments:

- 12,224 shipments are expected to be delivered by Motorbike, with Medium priority shipments accounting for 4,443 deliveries.

Regional Breakdown:

- Each transportation mode has its shipments distributed across different regions, including Lagos, Abuja, Port Harcourt, and more. The analysis helps identify key regional shipment trends and delivery performance.

Page 3: Customer Experience and Delivery Metrics

Key Insights:

Customer Type Breakdown:

- Business customers account for 35.99% of completed shipments, followed by Retail and International customers at 32.85% and 31.16% respectively.

Shipment Status:

- Shipment status metrics track various statuses such as In Transit, Ready for Pickup, Completed, etc.
- The priority levels help evaluate which shipments need immediate attention.

Satisfaction Level:

- Customer satisfaction is tracked for each shipment, with metrics like Very Satisfied, Neutral, and Unsatisfied reflecting the service quality.
- Shipment Value and Delivery Time are crucial factors in customer satisfaction, with high-value shipments and longer delivery times potentially impacting satisfaction levels.

Payment and Rating:

- The payment status (paid or pending) and delivery rating are tracked to correlate how the payment status impacts customer satisfaction.

CONCLUSION

This comprehensive Logistics Analysis Report allows stakeholders to get a clear picture of shipment performance, identify priority issues, and take action to improve logistics efficiency. The insights gained from this analysis can assist in enhancing shipment delivery timelines, optimizing transportation modes, and improving customer satisfaction.

KEY RECOMMENDATIONS

- Focus on improving the delivery times for Motorbike and Lorry shipments.
- Prioritize high-value and high-priority shipments to ensure they are delivered promptly.
- Use the regional insights to optimize shipment routes and carrier performance.

GOVERNANCE AND MAINTENANCE

ROLES AND PERMISSION

- **Executives** → View high-level KPIs (shipment value, satisfaction, overdue %).
- **Logistics Managers** → Use detailed dashboards (mode analysis, regional breakdowns) to make operational decisions.
- **Analysts** → Maintain DAX measures, monitor trends, and refine dashboards.
- **IT/Data Team** → Manage data refresh, security, and access control.

DATA REFRESH AND VERSION CONTROL

- **Refresh Schedule** → Daily refresh from source Excel/Power BI dataset.
- **Version Control** → Document changes to DAX measures and Power Query steps.
- **Audit Trail** → Keep logs of updates to ensure transparency.

DATA SENSITIVITY

- Shipment values and payment statuses are financially sensitive.
- Customer satisfaction scores are confidential and should not be shared externally.
- Access restricted by role (executives vs analysts vs IT).

KNOWN LIMITATIONS

- **Data Quality** → Missing or inconsistent dates can affect delivery time calculations.
- **Regional Coverage** → Some cities have fewer records, which may skew comparisons.
- **Payment Data** → Pending/overdue values may lag behind actual financial updates.
- **Satisfaction Scores** → Subjective and may not fully capture customer sentiment.

APPENDIX

GLOSSARY OF TERMS

- **Shipment Status** → Completed, In Transit, Ready for Pickup, Not Completed
- **Shipment Priority** → High, Medium, Low urgency levels
- **Customer Satisfaction Score** → Rating scale (1–10) reflecting customer feedback
- **Carrier Performance Rating** → Reliability score (1–5) for logistics partners
- **Shipment Value** → Monetary worth of a shipment
- **Payment Status** → Paid, Pending, Overdue

LIST OF DAX MEASURES

- **Performance KPIs** → % Completed, % In Transit, % Ready for Pickup, % Not Completed, Avg Delivery Day, Avg Satisfaction Score
- **Shipment Counts** → S_Completed, S_In Progress, S_In Transit, S_Not Completed, S_Ready for Pickup, Total Shipment
- **Financial Metrics** → Shipment Value, % Paid, % Pending, % Overdue
- **Mode & Priority Analysis** → Mode_Bus/Jets/Lorry/Motorbike, P_High/Medium/Low by mode
- **Dynamic Titles** → Location_Title, P_Title, Title measures for visuals
- **Trend Analysis** → MOM Completed, MOM In Progress, MOM Shipment Value, YOY Completed, YOY In Progress, YOY Shipment Value

(See Section 5 for full formulas and business meaning.)